TRANSPORTATION MODELS: OPTIMALITY TEST

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Some special items are produced at factories at Patna, Guwahati, Mumbai and Hyderabad. Distributors are located at Lucknow, Delhi, Jaipur, Chennai and Kolkata. Supply and demands of items are indicated in table. Note that factories have capacity to produce 18,000 items, but only 15,000 items are in demand. Transportation cost of each item is proportional to distance it travels. The road distances from factory to distribution centers (in km) are indicated in each cell. Your goal is to minimize the transportation cost.

Distributor Factory	Lucknow	Delhi	Jaipur	Chennai	Kolkata	Supply
Patna	610	1,060	1,180	1,490	580	3,000
Guwahati	1,020	1,910	1,946	2,700	1,018	6,000
Mumbai	1,370	1,420	1,150	1,330	2,050	4,000
Hyderabad	1,300	1,560	1,470	630	1,500	5,000
Demand	2,000	7,000	4,000	1,000	1,000	

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
Patna	610 **(2000)**	1060 **(1000)**	1180	1490	580	0	3000
Guwahati	1020	1910 **(6000)**	1946	2700	1018	0	6000
Mumbai	1370	1420	1150 **(4000)**	1330	2050	0	4000
Hyderabad	1300	1560	1470	630 **(1000)**	1500 **(1000)**	0 **(3000)**	5000
Demand	2000	7000	4000	1000	1000	3000	

The minimum total transportation cost by Northwest corner method

=610×2000+1060×1000+1910×6000+1150×4000+630×1000+1500×1000+0×3000 = 20,470,000.

Here, the number of allocated cells = 7, which is two less than to $m + n - 1 = 4 + 6 - 1 = 9$.
Hence, this solution is degenerate.
To resolve degeneracy, we make use of an artificial quantity(a=0). We have to make sure that there are 9 allocated cells. Hence, we allocate two artificial quantities (a=0).

The quantity (a) is assigned to that unoccupied cell, which will give zero opportunity cost i.e., $d_{ij} = c_{ij} - (u_i + v_j) = 0$

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
Patna	610(2000)	1060(1000)	1180	1490	580	0	3000
Guwahati	1020	1910(6000)	1946	2700	1018	0	6000
Mumbai	1370	1420 (a)	1150(4000)	1330	2050	0	4000
Hyderabad	1300	1560	1470 (a)	630(1000)	1500(1000)	0(3000)	5000
Demand	2000	7000	4000	1000	1000	3000	

Modified Distribution (MODI) Method

Find u_i and v_j for all occupied cells (i, j), where $u_i + v_j = c_{ij}$

Basic variable	(u, v) equation	Solution
x_{11}	$u_1 + v_1 = 610$	Set $u_1 = 0 \Rightarrow v_1 = 610$
x_{12}	$u_1 + v_2 = 1060$	$u_1 = 0 = v_2 = 1060$
x_{22}	$u_2 + v_2 = 1910$	$u_2 = 1910 - 1060 = 850$
x_{32}	$u_3 + v_2 = 1420$	$u_3 = 1420 - 1060 = 360$
x_{33}	$u_3 + v_3 = 1150$	$v_3 = 1150 - 360 = 790$
x_{43}	$u_4 + v_3 = 1470$	$u_4 = 1470 - 790 = 680$
x_{44}	$u_4 + v_4 = 630$	$v_4 = 630 - 680 = -50$
x_{45}	$u_4 + v_5 = 1500$	$v_5 = 1500 - 680 = 820$
x_{46}	$u_4 + v_6 = 0$	$v_6 = -680$

Find d_{ij} for all unoccupied cells (i,j), where $d_{ij} = c_{ij} - (u_i + v_j)$

Non basic variable	$d_{ij} = c_{ij} - (u_i + v_j)$
x_{13}	390
x_{14}	1540
x_{15}	–240
x_{16}	680
x_{21}	–440
x_{23}	306
x_{24}	1900
x_{25}	**–652 (minimum negative)**
x_{26}	170
x_{31}	400
x_{34}	1020
x_{35}	870
x_{36}	320
x_{41}	10
x_{42}	–180

Choose the minimum negative value from all d_{ij} (opportunity cost) = d_{25} = −652 and draw a closed path from corresponding cell (Guwahati, Kolkata).

Closed path is (Guwahati, Kolkata) → (Hyderabad, Kolkata) → (Hyderabad, Jaipur) → (Mumbai, Jaipur) → (Mumbai, Delhi) → (Guwahati, Delhi) → (Guwahati, Kolkata).

Iteration- 1

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply	u_i
Patna	610(2000)	1060(1000)	1180	1490	580	0	3000	$u_1 = 0$
Guwahati	1020	1910(6000–1000) –	1946	2700	1018 (1000) +	0	6000	$u_2 = 850$
Mumbai	1370	1420(0+1000) +	1150(4000–1000) –	1330	2050	0	4000	$u_3 = 360$
Hyderabad	1300	1560	1470 (0+1000) +	630(1000)	1500(1000–1000) –	0(3000)	5000	$u_4 = 680$
Demand	2000	7000	4000	1000	1000	3000		
v_j	$v_1 = 610$	$v_2 = 1060$	$v_3 = 790$	$v_4 = -50$	$v_5 = 820$	$v_6 = -680$		

After allocation, table becomes

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
Patna	610(2000)	1060(1000)	1180	1490	580	0	3000
Guwahati	1020	1910(5000)	1946	2700	1018(1000)	0	6000
Mumbai	1370	1420(1000)	1150(3000)	1330	2050	0	4000
Hyderabad	1300	1560	1470 (1000)	630(1000)	1500	0(3000)	5000
Demand	2000	7000	4000	1000	1000	3000	

Total transportation cost = 610 × 2000 + 1060 × 1000 +1910 × 5000 + 1018 ×1000 + 1420 ×1000 +1150 × 3000 +1470 × 1000 + 630 × 1000 + 0 × 3000 = 19,818,000

Find u_i and v_j for all occupied cells (i, j) where $u_i + v_i = c_{ij}$

Basic variable	(u, v) equation	Solution
x_{11}	$u_1 + v_1 = 610$	Set $u_1 = 0 \Rightarrow v_1 = 610$
x_{12}	$u_1 + v_2 = 1060$	$u_1 = 0 = v_2 = 1060$
x_{22}	$u_2 + v_5 = 1910$	$u_2 = 850$
x_{25}	$u_2 + v_5 = 1018$	$v_5 = 168$
x_{32}	$u_3 + v_2 = 1420$	$u_3 = 360$
x_{33}	$u_3 + v_3 = 1150$	$v_3 = 790$
x_{43}	$u_4 + v_3 = 1470$	$u_4 = 680$
x_{44}	$u_4 + v_5 = 630$	$v_4 = -50$
x_{46}	$u_4 + v_6 = 0$	$v_6 = -680$

Find d_{ij} for all unoccupied cells (i, j) where $d_{ij} = c_{ij} - (u_i + v_j)$

Non basic variable	$d_{ij} = c_{ij} - (u_i + v_j)$
x_{13}	390
x_{14}	1540
x_{15}	412
x_{16}	680
x_{21}	**–440 (minimum negative)**
x_{23}	306
x_{24}	1900
x_{26}	–170
x_{31}	400
x_{34}	1020
x_{35}	1522
x_{36}	320
x_{41}	10
x_{42}	–180
x_{45}	652

Choose the minimum negative value from all d_{ij} (opportunity cost) = d_{21} = –440 and draw a closed path from corresponding cell (Guwahati, Lucknow).

Closed path is (Guwahati, Lucknow) → (Patna, Lucknow) → (Patna, Delhi) → (Guwahati, Delhi) → (Guwahati, Lucknow).

Iteration-2

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply	u_i
Patna	610(2000–2000) –	1060(1000+2000) +	1180	1490	580	0	3000	$u_1 = 0$
Guwahati	1020(2000) +	1910(5000–2000) –	1946	2700	1018(1000)	0	6000	$u_2 = 850$
Mumbai	1370	1420(1000)	1150(3000)	1330	2050	0	4000	$u_3 = 360$
Hyderabad	1300	1560	1470(1000)	630(1000)	1500	0(3000)	5000	$u_4 = 680$
Demand	2000	7000	4000	1000	1000	3000		
v_j	$v_1 = 610$	$v_2 = 1060$	$v_3 = 790$	$v_4 = -50$	$v_5 = 168$	$v_6 = -680$		

After allocation, table becomes

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
Patna	610	1060(3000)	1180	1490	580	0	3000
Guwahati	1020(2000)	1910(3000)	1946	2700	1018(1000)	0	6000
Mumbai	1370	1420(1000)	1150(3000)	1330	2050	0	4000
Hyderabad	1300	1560	1470(1000)	630(1000)	1500	0(3000)	5000
Demand	2000	7000	4000	1000	1000	3000	

Total transportation cost = 18,938,000.

Find u_{ij} and v_{ij} for all occupied cells (i,j), where $u_i + v_j = c_{ij}$

Basic variable	(u, v) equation	Solution
x_{12}	$u_1 + v_2 = 1060$	Set $u_1 = 0 \Rightarrow v_2 = 1060$
x_{21}	$u_2 + v_1 = 1020$	$v_1 = 170$
x_{22}	$u_2 + v_2 = 1910$	$u_2 = 850$
x_{25}	$u_2 + v_5 = 1018$	$v_5 = 168$
x_{32}	$u_3 + v_2 = 1420$	$u_3 = 360$
x_{33}	$u_3 + v_3 = 1150$	$v_3 = 790$
x_{43}	$u_4 + v_3 = 1470$	$u_4 = 680$
x_{44}	$u_4 + v_5 = 630$	$v_4 = -50$
x_{46}	$u_4 + v_6 = 0$	$v_6 = -680$

Find d_{ij} for all for all unoccupied cells (i, j), where where $d_{ij} = c_{ji} - (u_i + v_i)$

Non basic variable	$d_{ij} = c_{ij} - \left(u_i + v_j\right)$
⋮	⋮
x_{42}	−180 **(minimum negative)**
x_{45}	652

Choose the minimum negative value from all d_{ij} (opportunity cost) = $d_{42} = -180$ and draw a closed path from corresponding cell ((Hyderabad, Delhi).

Closed path is (Hyderabad, Delhi) → (Mumbai, Delhi) → (Mumbai, Jaipur) → (Hyderabad, Jaipur) → (Hyderabad, Delhi).

Iteration-3

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply	u_i
Patna	610	1060(3000)	1180	1490	580	0	3000	$u_1 = 0$
Guwahati	1020(2000)	1910(3000)	1946	2700	1018(1000)	0	6000	$u_2 = 850$
Mumbai	1370	1420(1000–1000) −	1150(3000+1000) +	1330	2050	0	4000	$u_3 = 360$
Hyderabad	1300	1560 (1000) +	1470(1000–1000) −	630(1000)	1500	0(3000)	5000	$u_4 = 680$
Demand	2000	7000	4000	1000	1000	3000		
v_j	$v_1 = 170$	$v_2 = 1060$	$v_3 = 790$	$v_4 = -50$	$v_5 = 168$	$v_6 = -680$		

After allocation, table becomes

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
Patna	610	1060(3000)	1180	1490	580	0	3000
Guwahati	1020(2000)	1910(3000)	1946	2700	1018(1000)	0	6000
Mumbai	1370	1420	1150(4000)	1330	2050	0	4000
Hyderabad	1300	1560 (1000)	1470	630(1000)	1500	0(3000)	5000
Demand	2000	7000	4000	1000	1000	3000	

Total transportation cost = 18,758,000.

Here, the number of allocated cells = 8, which is one less than $m + n - 1 = 4 + 6 - 1 = 9$.
Hence, this solution is degenerate.
To resolve degeneracy, we allocate one artificial quantity (a=0). The quantity a is assigned to that unoccupied cell, which will give zero opportunity cost i.e., $d_{ij} = c_{ij} - (u_i + v_j) = 0$

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
Patna	610	1060(3000)	1180	1490	580	0	3000
Guwahati	1020(2000)	1910(3000)	1946	2700	1018(1000)	0	6000
Mumbai	1370	1420	1150(4000)	1330	2050	0	4000
Hyderabad	1300	1560 (1000)	1470 (a)	630(1000)	1500	0(3000)	5000
Demand	2000	7000	4000	1000	1000	3000	

Find u_{ij} and v_{ij} for all occupied cells (i,j), where $u_i + v_j = c_{ij}$

Basic variable	(u, v) equation	Solution
x_{12}	$u_1 + v_2 = 1060$	Set $u_1 = 0 \Rightarrow v_2 = 1060$
x_{21}	$u_2 + v_1 = 1020$	$v_1 = 170$
x_{22}	$u_2 + v_2 = 1910$	$u_2 = 850$
x_{25}	$u_2 + v_5 = 1018$	$v_5 = 168$
x_{33}	$u_3 + v_3 = 1150$	$u_3 = 180$
x_{42}	$u_4 + v_2 = 1560$	$u_4 = 500$
x_{43}	$u_4 + v_3 = 1470$	$v_3 = 970$
x_{44}	$u_4 + v_5 = 630$	$v_4 = 130$
x_{46}	$u_4 + v_6 = 0$	$v_6 = -500$

Find d_{ij} for all for all unoccupied cells (i, j), where where $d_{ij} = c_{ji} - (u_i + v_i)$

Non basic variable	$d_{ij} = c_{ij} - (u_i + v_j)$
x_{11}	440
⋮	⋮
x_{26}	–350 **(minimum negative)**
⋮	⋮
x_{45}	832

Choose the minimum negative value from all d_{ij} (opportunity cost) = $d_{26} = -350$ and draw a closed path from corresponding cell (Guwahati, D_{dummy}).

Closed path is (Guwahati, D_{dummy}) → (Hyderabad, D_{dummy}) → (Hyderabad, Delhi) → (Guwahati, Delhi) → (Guwahati, D_{dummy}).

Iteration-4

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply	u_i
Patna	610	1060(3000)	1180	1490	580	0	3000	$u_1 = 0$
Guwahati	1020(2000)	1910(3000–3000) −	1946	2700	1018(1000)	0 (3000) +	6000	$u_2 = 850$
Mumbai	1370	1420	1150(4000)	1330	2050	0	4000	$u_3 = 180$
Hyderabad	1300	1560(1000+3000) +	1470 (0)	630(1000)	1500	0(3000–3000) −	5000	$u_4 = 970$
Demand	2000	7000	4000	1000	1000	3000		
v_j	$v_1 = 170$	$v_2 = 1060$	$v_3 = 970$	$v_4 = 130$	$v_5 = 168$	$v_6 = -500$		

After allocation, table becomes

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
Patna	610	1060(3000)	1180	1490	580	0	3000
Guwahati	1020(2000)	1910	1946	2700	1018(1000)	0 (3000)	6000
Mumbai	1370	1420	1150(4000)	1330	2050	0	4000
Hyderabad	1300	1560 (4000)	1470 (0)	630(1000)	1500	0	5000
Demand	2000	7000	4000	1000	1000	3000	

Total transportation cost = 17,708,000.

Here, the number of allocated cells = 8, which is two less than to $m + n - 1 = 4 + 6 - 1 = 9$.
Hence, this solution is degenerate.
To resolve degeneracy, we allocate one artificial quantity (a=0) in cell (Hyderabad, D_{dummy}).

	Lucknow	Delhi	Jaipur	Chennai	Kolkata	D_{dummy}	Supply
Patna	610	1060(3000)	1180	1490	580	0	3000
Guwahati	1020(2000)	1910	1946	2700	1018(1000)	0 (3000)	6000
Mumbai	1370	1420	1150(4000)	1330	2050	0	4000
Hyderabad	1300	1560 (4000)	1470 (0)	630(1000)	1500	0 (a)	5000
Demand	2000	7000	4000	1000	1000	3000	

Find u_i and v_j for all occupied cells (i,j) where $u_i + v_j = c_{ij}$

Basic variable	(u, v) equation	Solution
x_{12}	$u_1 + v_2 = 1060$	Set $u_1 = 0 \Rightarrow v_2 = 1060$
x_{21}	$u_2 + v_1 = 1020$	$v_1 = 520$
x_{25}	$u_2 + v_5 = 1018$	$v_5 = 500$
x_{26}	$u_2 + v_6 = 0$	$u_2 = 168$
x_{33}	$u_3 + v_3 = 1150$	$u_3 = 180$
x_{42}	$u_4 + v_2 = 1560$	$u_4 = 500$
x_{43}	$u_4 + v_3 = 1470$	$v_3 = 970$
x_{44}	$u_4 + v_5 = 630$	$v_4 = 130$
x_{46}	$u_4 + v_6 = 0$	$v_6 = -500$

Find d_{ij} for all unoccupied cells (i, j), where $d_{ij} = c_{ij} - (u_i + v_j)$

It can be shown that $d_{ij} \geq 0$ for all unoccupied cells (i, j). Hence, the solution is optimal.

Following table summarizes the optimum solution:

Source	**Destination**	**Transportation cost**
Patna	Delhi	1060×3000 = 3,180,000
Guwahati	Lucknow	1020×2000 = 2,040,000
Guwahati	Kolkata	1018×1000 = 1,018,000
Mumbai	Jaipur	1150×4000 = 4,600,000
Hyderabad	Delhi	1560×4000 = 6,240,000
Hyderabad	Chennai	630×1000 = 630,000
Minimum total transportation cost		Rs. **17,708,000.**

$u_1 + v_1 = 80$ $u_1 = 0 \Rightarrow v_1 = 80$ $u_1 + v_2 = 21$ $v_2 = 21 - 0 = 21$ $u_2 + v_1 = 100$ $u_2 = 100 - 80 = 20$

$u_3 + v_2 = 8$ $u_3 = 8 - 21 = -13$

	$Market_1$	$Market_2$	Supply
$Plant_1$	80**(800)**	21**(200)**	1000
$Plant_2$	100**(1500)**	108	1500
$Plant_3$	102	8**(1200)**	1200
Demand	2300	1400	

Basic variable	(u, v) equation	Solution
x_{11}	$u_1 + v_1 = 80$	Set $u_1 = 0 \Rightarrow v_1 = 80$
x_{21}	$u_2 + v_1 = 100$	$u_2 = 100 - 80 = 20$
x_{12}	$u_1 + v_2 = 21$	$v_2 = 21 - 0 = 21$
x_{32}	$u_3 + v_2 = 8$	$u_3 = 8 - 21 = -13$